

7 Green’s Function

ps: To solve forced equation. 即 PDE RHS ≠ 0

Green's Function: For PDE $L u(x, t) = F(x, t)$ where L is linear differential operation with BCs and ICs.

- 1. Consider $G(x, t; \xi, \tau)$ satisfying $L G(x, t; \xi, \tau) = \delta(x - \xi)\delta(t - \tau)$ with BCs and ICs.
- 2. Solve $G \rightarrow G(x, t; \xi, \tau)$.
- 3. Then $u(x, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} G(x, t; \xi, \tau) F(\xi, \tau) d\xi d\tau$.

ps: 可以拓展到更多的空间维度, e.g. $G(x, y, t; \xi, \eta, \tau) \rightarrow L G(x, y, t; \xi, \eta, \tau) = \delta(x - \xi)\delta(y - \eta)\delta(t - \tau)$

- How to solve: $L G(x, t; \xi, \tau) = \delta(x - \xi)\delta(t - \tau)$ with BCs and ICs. ↓

- a. **Fourier Transform** (infinite space domain (x, t) , [2 dimensions])
 - 1. Take Fourier transform in $x \rightarrow$ Get ODE $\rightarrow$ Solve ODE with $t \neq \tau \rightarrow$ Get $\overline{G}$ split by $t < \tau$ and $t > \tau$.
 - 2. ICs $\rightarrow$ Get some unknown constants in $\overline{G}$.
 - 3. Apply **Jump Condition** to ODE, $\int_{\tau-}^{\tau+} \cdot dt \rightarrow$ Get unknown constant.
 - 4. Inverse Fourier $\rightarrow G(x, t; \xi, \tau)$.
- b. **Separation of Variables** (finite space domain (x))
 - 1. Need homogeneous BCs $\rightarrow$ Assume $t \neq \tau$ and $x \neq \xi \rightarrow$ Seperation of variables $\rightarrow$ Get λ_n and $X_n(x)$.
 - 2. Assume $G = \sum_n g_n(t) X_n(x) \rightarrow$ Substitute into PDE $L G = \delta(x - \xi)\delta(t - \tau)$.
 - 3. Use orthogonality of $X_n \rightarrow$ Get $\delta(x - \xi) = \sum_n d_n X_n(x)$ and find d_n .
 - 4. 通过前面 substitute 得到的对比 terms $\rightarrow$ Get ODE for $g_n(t) \rightarrow$ Solve ODE split by $t < \tau$ and $t > \tau$.
 - 5. ICs $\rightarrow$ Get some unknown constants in $g_n(t)$. $\rightarrow$ **Continuity** at $t = \tau \rightarrow$ some unknown constants.
 - 6. Apply **Jump Condition** to ODE, $\int_{\tau-}^{\tau+} \cdot dt \rightarrow$ Get unknown constant $\rightarrow$ Get $g_n(t)$ and we solve it :)
 - Remark: 可以拓展为更多空间维度 $\rightarrow$ Separation of variables 后留空间不算到最后假设来找.
- c. **High Variables / Laplace Transform**

Remark: 如 $F(x, t)$ 给具体函数 $\rightarrow$ 考虑 homogeneous PDE $\rightarrow$ Eigenfunction $X_n(x) \rightarrow u = \sum_n g_n(t) X_n(x)$
RHS 含 x 的写为 $\sum_n d_n X_n(x) \rightarrow$ 比较系数 $\rightarrow$ 得到 $g_n(t)$ 的 ODE $\rightarrow$ Solve ODE $\rightarrow$ 最后得到 $u(x, t)$

8 Laplace Transform Table

$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$	t^n	$\frac{n!}{s^{n+1}}$	$f'(t)$	$s\overline{f}(s) - f(0)$
e^{at}	$\frac{1}{s-a}$	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$\int_0^t f(\tau) d\tau$	$\frac{1}{s}\overline{f}(s)$
$\sin(at)$	$\frac{a}{s^2+a^2}$	$t^n \sin(at)$	$\frac{n!a}{(s^2+a^2)^{n+1}}$	$e^{at} f(t)$	s-shift $\overline{f}(s-a)$
$\cos(at)$	$\frac{s}{s^2+a^2}$	$t^n \cos(at)$	$\frac{n!s}{(s^2+a^2)^{n+1}}$	$f(t-a)u_a(t)$	t-shift $e^{-as}\overline{f}(s)$
$\sinh(at)$	$\frac{a}{s^2-a^2}$	$t^n \sinh(at)$	$\frac{n!a}{(s^2-a^2)^{n+1}}$	$\int_0^t f(t-\tau)g(\tau) d\tau$	Convolution $\overline{f}(s) \cdot \overline{g}(s)$
$\cosh(at)$	$\frac{s}{s^2-a^2}$	$t^n \cosh(at)$	$\frac{n!s}{(s^2-a^2)^{n+1}}$	$f^n(t): s^n \overline{f}(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$	
$u_b(t)$	$\frac{e^{-bs}}{s}$	$\delta(t-b)$	e^{-bs}	$f * g = \int_0^t f(t-\tau)g(\tau) d\tau$	

ps: $u_b(t) = H(t-b) = \begin{cases} 0, & t < b \\ 1, & t \geq b \end{cases}$ $\int_{-\infty}^{\infty} \delta(t-b)\phi(t)dt = \int_b^{b+} \delta(t-b)\phi(t)dt = \phi(b)$

9 Appendix II

Useful Tools: $\int_0^L \sin^2(ax)dx = \frac{L}{2} - \frac{\sin(2aL)}{4a}$ $\int_0^L \cos^2(ax)dx = \frac{L}{2} + \frac{\sin(2aL)}{4a}$
 $\int_0^L \sin(ax)\cos(ax)dx = \frac{\sin^2(aL)}{2a}$